PASCAL'S WAGER

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https://www.princeton.edu/~grosen/puc/phi203/Pascal.html

Pascal concedes that a belief in God's existence cannot be supported by argument or evidence, but maintains that religious belief is rationally required nonetheless.

Theoretical vs. Practical Rationality

The argument tacitly exploits the distinction between theoretical rationality and practical rationality. (These are our terms, not Pascal's.) Theoretical rationality is (roughly) a matter of evidential and argumentative support. Your belief in God is rational in the theoretical sense just in case the balance of evidence and argument supports the truth of the proposition that God exists. Your belief in God is practically rational, on the other hand, if it is in your interest for you to hold it.

To illustrate this contrast, consider the following sort of case. The evidence suggests that X does not love you. X ignores you at parties; X won't return your calls, etc. And yet you are so desperately in love with X that you would not be able to function if you came to believe that your love was unreciprocated. You would not be able to get out of bed or get to work. Your life would deteriorate in every meaningful respect. Now ask: is it rational for you to believe that X

loves you? In one sense, the answer is clearly "no": all the evidence is against it. But in another sense, the answer is clearly "yes". You may care very much about theoretical rationality, believing only what can be supported by argument. But you may well care more about being able to get through the day. And if you do, it may well be practically rational for you to believe that X loves you. This belief, more than any other, serves your interests in the circumstances.

The example shows that theoretical rationality and practical rationality can clash, and this raises a question. When the two sorts of rationality conflict, what is it most rational to do? All things considered, should you believe that X loves you or not? When we are told that the belief is theoretically irrational but practically rational, we are given no answer to this seemingly more fundamental question. Indeed, we seem to have been told that the question is simply ambiguous, and so need not have a single answer. If the "should" that figures in the question is the "should" of theoretical rationality, then you should not believe; if it is the "should" of practical rationality, then you should. It would be interesting to know if there were a third, more general sense of the word "should" that permits an unambiguous formulation of the question, and if so, what its answer should be.

We are not going to pursue this interesting theoretical question. We note the distinction because it permits us to state Pascal's conclusion in non-paradoxical terms. The conclusion of the argument is that while belief in God may not be required by the norms or requirements of theoretical rationality, it is required by the norms of practical rationality. The case is not one of a straightforward clash: the norms of theoretical rationality are (as it were) silent about God's existence on Pascal's view. If we attend only to reason and argument, it is conceivable that God exists, but also conceivable that he does not. Pascal's approach is to invoke practical rationality in order to break the tie. It would be interesting to know whether Pascal explicitly

entertains the possibility that theoretical reason and practical reason might clash on this matter, and if so, whether he maintains that we should—all things considered—adhere to the requirements of practical rationality in this case.

A Crash Course in Decision Theory

The Wager argument tacitly exploits some principles of practical rationality that have come to be enshrined in an explicit theory, sometimes called "decision theory" or "rational choice" theory. Considered as a descriptive account of human choice, decision theory and its closely related partner, game theory, have come to serve as the foundation for modern economics. For our purposes, however, it is unnecessary to claim that people actually do for the most part act as the theory says they should. For us it is enough to treat the theory as a normative account: an account of how a rational agent "ought" to act in any given circumstance.

The theory is an attempt to make precise the thought that a rational agent ought always to perform the act that best promotes his interests as he conceives them. These interests need not be "selfish". If you care very much about helping the poor, then it will be in your interest to try to help them. So, while the theory may be understood as an account of self-interested action, self-interest must be understood broadly. To act self-interestedly is to pursue what you regard as valuable or worthwhile.

Before I state the theory explicitly—and I will only state the most elementary part of it—let me give you an example to show how it is supposed to work. Suppose you are going over to a friend's house for dinner. It's your job to bring the wine, and you can bring either white or red. The trouble is that you don't know what sort of food you'll be having. For all you know, it could be chicken or it could be

beef. If the main course is going to be beef, you strongly prefer red wine to white—white wine with beef is an abomination. If it's chicken, on the other hand, you prefer white wine to red. But here your preference is not so strong: red wine with chicken is a perfectly tolerable second best. We might represent your preferences then as follows.

	they serve beef	they serve chicken
bring red wine	10	5
bring white wine	0	10

The numbers here represent the strength of your preference, and obviously they are somewhat arbitrary. Think of them as representing units of "goodness" from your point of view. That red wine with beef gets a higher score than red wine with chicken shows that from your point of view the first outcome is better than the second.

Now before we can decide what you ought to do, we need to say something about the probabilities or likelihoods that you attach to the two possible "states of nature". Suppose first that you think your friends are just as likely to make beef as chicken. Then we say that for you P(*beef*) = P(*chicken*) = 0.5

In general, probabilities are measured by numbers between 0 and 1. If you attach probability 1 to a proposition, you are certain that it is true; if you attach probability 0 to it, you are certain that it is false. If attach probability 2/3 to a proposition, that means that you think that it is twice as likely to be true as it is to be false: your odds in favor of the proposition are 2:1. And so on.

Given this assignment of probabilities, we can calculate the *expected utility* of each of the acts available to you. This is the sum of

the good you can expect to achieve by performing a given act given a certain state of nature, weighted by the probability you attach to that state. In this case:

EU(*red*) = 0.5 (10) + 0.5(5) = 7.5

EU(*white*) = 0.5 (0) + 0.5(10) = 5

The central principle of the theory of rational choice may then be invoked: *A rational agent should perform the act with the greatest expected utility*. In this case, the principle implies—plausibly enough—that you should bring a bottle of red wine.

Just to get a sense of how expected utility depends on probabilities as well as values, suppose now that your friends are great chicken lovers. You've been to dinner at their house 100 times, and on 90 occasions they have served chicken rather than beef. On this basis you conclude that P(*beef*) = 0.1, P(*chicken*) = 0.9. The expected utilities are then as follows:

EU(*red*) = 0.1 (10) + 0.9(5) = 5.5

EU(*white*) = 0.1 (0) + 0.9 (10) = 9

In these circumstances, you should bring white wine.

The general rule for calculating expected utilities can be stated as follows:

$$Expected\ Utility(A) = \sum_{j=1}^{N} P(S_j) \cdot utility(A|\, S_j)$$

In words: where there are N possible states of nature S_1 through S_N—states that might obtain for all you know—the expected utility of a given act A is the sum over all states of nature of the utility of the act in a given state of nature multiplied by the probability that that state of nature obtains.

The rule is not entirely uncontroversial, and you might try to imagine ordinary (or extraordinary) cases where it gives the wrong

result. For now, however, we will assume that something like this principle is in place.

The Wager

What does this have to do with the rationality of religious belief? Just this. Suppose that you've concluded that theoretical reason cannot settle the question of God's existence one way or the other. You might then be persuaded by the following argument. There are two possibilities: either God exists or he does not. And there are two relevant actions I can take: I can believe or I can fail to believe. (This last option covers both the case of positive disbelief and the case of agnosticism.) So we can begin to represent our decision problem as follows:

	God exists	God doesn't exist
believe that God exists		
don't believe that God exists		

To flesh out the problem, we need to assign values or utilities to the various acts in each of the various states of nature and probabilities to the various states. For simplicity, let's assume that we are perfectly agnostic at the start: P(*exists*) = P(*doesn't exist*) = 0.5

As for utilities, we know something about how they should be represented. (Mark 16:16: "He that believeth and is baptized shall be saved; but he that believeth not shall be damned"). If we believe—and this means genuine belief, and not merely feigned belief or insincere assertion—and God exists, then we shall be infinitely happy: our happiness will be infinite in duration, and it will be unsurpassable in

degree. If we believe and God does not exist, then we will have been wrong about an important matter. We will have missed out on some of the secular pleasures we might otherwise have enjoyed; but we will also have been comforted (however falsely) by our faith and we will not ever have suffered from knowing that we were wrong. So a religious life in a Godless world is a mixed blessing, with which we should associate some finite utility or disutility. Pascal believes that this utility is positive, and also that it is greater than the utility of a secular life in a Godless world. But that is controversial, so let's not make this assumption. Somewhat arbitrarily, let us suppose that the value of believing in a Godless world is -100; whereas the value of disbelieving in a Godless world is much greater, say 1000. (As we shall see, the numbers don't matter at all as long as they are finite.) Finally, it seems clear that if God does exist and we willfully choose not to believe that he does, we must suffer some very serious penalty. In Pascal's representation, the penalty is infinite disutility -- the agonies of hell. On this assumption, the payoff matrix looks like this:

	God exists	God doesn't exist
believe that God exists	∞	-100
don't believe that God exists	-∞	1000

("∞" is the sideways 8. It stands for "infinity.")

Given our initial assignment of probabilities, we then have the following expected utilities.

EU(believe) = 0.5 (∞) + 0.5 (-100) = ∞

EU(don't believe) = 0.5 (-∞) + 0.5 (1000) = -∞

In short: a rational agent will do what he can to believe that God exists.

Of course it will strike you immediately that the numbers we have used in this calculation are arbitrary. But it should be equally obvious that the particular numbers don't matter. Even if you begin by assuming that the probability of God's existence is miniscule—say, 1 in 1,000,000—the argument will still go through. The assumptions that underlie the Wager are simply as follows:

- The utilities associated with *believe* and *don't believe* given that God exists are ∞ and $-\infty$ respectively.
- The utilities in the case that God doesn't exist are finite.
- Your probability that God exists is greater than zero. (I.e., you are not initially certain that God does not exist.)
- An infinite number is greater than any finite number.
- The addition of a finite quantity (positive or negative) to an infinite quantity is still infinite.
- Multiplying an infinite quantity by any positive finite number, even a fraction, yields an infinite quantity.

If we make these assumptions, you can plug in any finite values you like for the utilities in the *doesn't exist* column and for the probability of God's existence. Given the principle that rational agent's seek to maximize expected utility, the Wager will still go through.

The Problem of Many Gods

The most powerful objection to Pascal's Wager is the so-called "Many Gods" objection. The objection concedes that it makes sense to think of the problem as a problem about practical rationality, and it does not quarrel with Pascal's assumption that the Christian God rewards belief and punishes disbelief as we have suggested. The objection is rather that Pascal has failed to take account of certain possibilities.

One possibility is that the Christian God exists, a God who rewards those who believe in him and punishes those who do not believe. Another possibility is that there is no God at all. Pascal explicitly asserts that these are the only two possibilities. But that is not true. Leaving aside the manifold religions and sects with their manifold conceptions of God and his attitudes towards religious belief, we can certainly imagine a being whom I will call the *Perverse God*. The Perverse God rewards those who believe only what can be established on the basis of objectively compelling evidence and argument. He is the God of theoretical rationality, whose wrath is reserved for those who are intellectually lazy or sloppy, but also for those who would bring the canons of practical rationality to bear on epistemological matters. Since there are no compelling theoretical grounds for believing that any sort of God exists, the Perverse God punishes those who believe (for whatever reason) and rewards those who were sensible enough to reserve their judgment. (If we suppose that there are no compelling theoretical grounds for atheism, the Perverse God will punish atheists and theists alike.) We may suppose that his punishments and rewards are also infinite—infinite happiness for those who suspend judgment, infinite misery for those who do not. Given this possibility, the payoff matrix to consider is the following:

	P = *p*	P = *q*	P = *r*
	Christian God exists	perverse God exists	no God exists
believe that a God exists	∞	-∞	-100
suspend judgment as to the existence of God	-∞	∞	500
deny the existence of God	-∞	-∞	1000

And now it should be obvious how the expected utility calculations ought to go. Let p, q, and r be the probabilities that you assign to the three possible states respectively, and assume that they are all neither 0 nor 1. We then have:

EU(*believe*) = $p(\infty) + q(-\infty) + r(-100) = r(-100)$*

EU(*suspend*) = $p(-\infty) + q(\infty) + r(500) = r(500)$

EU(*don't believe*) = $p(-\infty) + q(-\infty) + r(1000) = -\infty$

Given this more detailed, way of setting up the problem, it appears that the rational agent ought to suspend judgment.

But of course this is somewhat arbitrary. For as soon as we introduce the possibility of the Perverse God, it turns out that the

* This is not obviously correct. The infinite is like the finite in some mathematical respects, but very different in others. There is more than one way to make this line of thought mathematically rigorous. Some ways will validate this crude line of reasoning. Others won't. If you know something about transfinite arithmetic, this might be an interesting topic to pursue in a paper.

rational choice depends crucially on the finite numbers we assign in the last column. But these numbers were soft. It might be that the pleasures and comforts of belief outweigh the anxiety of agnosticism. In that case, perhaps these numbers should be reversed, in which case we would have a different recommendation. The general point is that any argument for the existence of God that depends crucially on how these finite numbers are assigned is not to be taken seriously unless much more can be said about what these numbers ought to be.

But that is not the only problem. These calculations work out as they do because we have included only two theistic possibilities. If we were to add others—a Jealous God who rewards only those who believe in him and call him by name, punishing the rest, including the Christian believers, a shy God who rewards only those who believe in some other God while punishing atheists, agnostics, and those who believe in him, etc.—we can obviously rig the matrix to yield any answer we like. A complete formulation of the problem would have to take into account each of these relevant possibilities, adding a new column for each sort of God. And it should be obvious that we have no idea what such a complete representation would look like, nor even whether the idea makes any sense. In view of all this, it seems premature—to say the least—to place any confidence at all in the argument as Pascal presents it.

Hajek's Objection

The Many Gods objection is quite serious. You might consult the discussion by Lycan and Schlesinger in the textbook for some possible responses.[†] But I will now suggest that even if this problem were somehow solved, Pascal's Wager would still be invalid. I first heard

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[†] Lycan & Schlesinger, You Bet Your Life: Pascal's Wager Defended. In J. Feinberg (ed.), *Reason and Responsibility*. Wadsworth (1988).

this problem from Alan Hajek, who teaches philosophy at Cal Tech. Apparently he was not the first to discover it; but it was discovered only very recently, and if nothing else this should convince you that there are still new things to say about old and much-discussed philosophical arguments.

So far we have been supposing that the practical problem we face is to decide whether or not to believe that God exists. But as Pascal freely admits, belief is not simply a matter of the will. We cannot simply choose to believe anything. We can decide to try to believe, or to take steps that are likely to bring about belief. We can, as Pascal says, attend Mass, take holy water, pray, meditate, read the Bible, and so on. But we cannot simply choose to believe that God exists. So the acts we really ought to be considering are *try to believe* and *don't try to believe*.

But this makes an enormous difference to how we understand the decision problem. Suppose first that God exists, and suppose that you try to believe. Before we took it for granted that the upper left hand corner of the payoff matrix should contain an infinite payoff, since the divine reward for believing is infinite felicity. But now this is not so clear. If you try to believe there is some probability that you will succeed, in which case you will get the infinite benefit; but there is also some probability that you will fail, in which case your will pay the infinite penalty. Suppose that p is the probability of succeeding in your most earnest attempt to believe, and $1 - p$ is the probability of failing. Then the expected utility of *try to believe*, given that God exists, is

$$p(\infty) + (1- p)(-\infty) = 0,$$

assuming as before that p is neither 0 nor 1. (See the footnote above for a caveat about this sort of reasoning.)

But now, here's the kicker: even if you *don't try* to believe, there is some chance that you will wind up believing. As with Saul on the road to Damascus, belief sometimes comes to those who have made

no effort whatsoever to attain it. So suppose once again that God exists, and consider the expected utility of *don't try*. There is some miniscule probability p^* that you will wind up believing anyway, and some much larger probability $1 - p^*$ that you will remain a non-believer. The expected utility of the act is then given by

$$p^*(\infty) + (1 - p^*)(-\infty) = 0$$

What this means is that even if God exists, we have no more reason to try to believe than not try. It is true that trying to believe increases the probability that we will secure infinite felicity and avoid infinite misery. But given the principles about infinity we assumed above, especially the principle that says that an infinite quantity divided by any finite quantity is still infinite, we seem forced to say that any shot at an infinite gain has infinite expected utility. The probability of success does not matter.

What this shows is first that the decision matrix must be more complicated than we have let on. The value of the act of trying to believe depends on two independent factors: the existence of God and the success of the attempt. The value of not trying likewise depends on two factors: the existence of God and the probability of your coming to believe without trying. As a first approximation, we might represent the problem as follows:

	God exists; you believe if you try; you believe if you don't try	God exists; you believe if you try; you don't believe if you don't try	God exists; you don't believe if you try; you believe if you don't try	God exists; you don't believe if you try; you don't believe if you don't try	God does not exist; you believe if you try; you believe if you don't try	God does not exist; you believe if you try; you don't believe if you don't try	God does not exist; you don't believe if you try; you believe if you don't try	God does not exist; you don't believe if you try; you don't believe if you don't try
try to believe	∞	∞	$-\infty$	$-\infty$	-100	-100	-500 (?)	- 500
don't try to believe	∞	$-\infty$	∞	$-\infty$	-100	1000	-100	1000

I will leave it to you to work out what the rational act might be under these assumptions. But as we have set it up here, it should be clear that if the infinite utilities cancel out, the answer will depend on the finite numbers on the right hand side of the diagram. And as I have stressed, those numbers are soft. Given the values I have penciled in, the act of not trying to believe will be favored. It "weakly dominates" the act of trying to believe, in the sense that no matter what column you're in (on the right hand side) you do at least as well by not trying as by trying, and sometimes you do better. But a different choice of numbers would yield a different result. So at this point it seems clear that we can have no confidence in any calculation of this sort. Pascal's Wager is convincing only if the finite utilities are swamped. But here they are not. So the Wager should not move us.

This is not the last word on the Wager, of course. If you are not put off by the (admittedly somewhat fishy) mathematics that this objection involves, you might try to work out a response.